Decision Making in Machines / Computers

Artificial Intelligence – Reading Material

Unit 1 (Three Chapters)

What you will learn in this chapter:

- The difference between **Automated** and **Autonomous** systems, and between **Deterministic** and **Probabilistic** behaviour.
- How **humans** and **machines** make decisions – and why one is called subjective while the other is objective.
- The basics of **Machine Learning**, the role of data, and the steps a machine follows to "learn."

1 Decision Making in Machines / Computers

Every day, you make decisions – what to wear, which route to take to school, whether to carry an umbrella. Machines and computers make decisions too: a washing machine decides when to stop spinning, a traffic signal decides when to turn red, and a self-driving car decides when to brake. But *how* a decision is made, and *how well* the machine can handle new or unexpected situations, differs enormously from one system to another. This chapter builds the foundation for understanding how machines "think" and decide.

1.1 Automated versus Autonomous Systems

1.1.1 Automated Systems

An **Automated System** is a system that performs a task by strictly following a set of **pre-defined instructions or rules** written by a human programmer. It does exactly what it has been programmed to do – nothing more, nothing less. It does not think, adapt, or handle situations it was not designed for.

Definition

An **Automated System** is a machine or program that carries out a fixed, pre-programmed sequence of actions in response to specific inputs, without adapting its behaviour based on new experience.

Example – Automated Systems

- A **washing machine** runs through wash, rinse, and spin cycles for a fixed duration, regardless of how dirty the clothes actually are.
- A **traffic signal** that changes from red to green to yellow after a fixed number of seconds, irrespective of the actual traffic on the road.
- An **automatic door** that opens whenever a sensor detects motion within a certain range.

1.1.2 Autonomous Systems

An **Autonomous System**, on the other hand, can **sense its environment, analyse the situation, and decide its own course of action** – often without any direct human control. It uses sensors, data, and intelligent algorithms (frequently powered by AI) to adapt its behaviour to changing conditions.

Definition

An **Autonomous System** is a system capable of perceiving its environment through sensors and making independent decisions to achieve a goal, adapting its actions as conditions change, with minimal or no human intervention.

Example – Autonomous Systems

- A **self-driving car** that senses obstacles, pedestrians, and traffic, and decides on its own when to brake, turn, or accelerate.
- A **robot vacuum cleaner** that maps a room, detects obstacles, and changes its cleaning path accordingly.
- A **drone** that autonomously adjusts its flight path to avoid a sudden obstacle.

Basis	Automated System	Autonomous System
Behaviour	Follows fixed, pre-set rules	Senses, analyses, and decides on its own
Adaptability	Cannot adapt to new/unseen situations	Adapts its actions to changing environments
Human control	Needs instructions for every situation	Works with minimal human intervention
Intelligence involved	Little or none	Uses AI / sensors / data-driven reasoning
Example	Washing machine, traffic light	Self-driving car, robot vacuum cleaner

1.1.3 Deterministic versus Probabilistic Behaviour

Closely linked to this idea is *how predictable* a system's output is. This depends on whether the system behaves in a **Deterministic** or a **Probabilistic** manner.

Definition – Deterministic and Probabilistic

A **Deterministic system** always produces the *same, exact output* for a given input – there is no uncertainty involved.

A **Probabilistic system** produces outputs based on the *likelihood or probability* of an outcome; the result may vary and involves an element of uncertainty, usually because it is learned from data.

How the two ideas connect

- Automated systems are almost always **deterministic** – since they follow fixed rules, the same input always gives the same output.

- Autonomous systems are usually **probabilistic** – since they rely on sensor data and learned patterns, the exact same situation may lead to slightly different decisions based on probability and confidence levels.

Example – Deterministic vs Probabilistic

A simple calculator is **deterministic**: 7×8 will *always* give 56.
A weather-prediction AI is **probabilistic**: it may say "there is an *80% chance* of rain today," which is a statement of likelihood, not a certainty.

Think About It

Is an elevator that stops at floors when a button is pressed an automated system or an autonomous system? Is it deterministic or probabilistic? Discuss with a classmate.

Quick Recap

- Automated = fixed instructions, no adaptability, usually deterministic.
- Autonomous = senses, decides, adapts, usually probabilistic.
- Deterministic = same input → same output always.
- Probabilistic = output based on likelihood; may vary.

1.2 Decision Making

Decision making is the process of choosing an action from among several possibilities. Both humans and machines make decisions constantly – but the *way* they arrive at a decision is fundamentally different.

1.2.1 Human Decision Making – Subjective

Human beings make decisions based on **feelings, past experiences, personal values, intuition, and context**. Two different people, given the exact same situation, may arrive at two completely different decisions. This is why human decision making is called **subjective**.

Definition

Subjective decision making is decision making that is influenced by personal opinions, emotions, experiences, and perspective, and can therefore differ from person to person even when the situation is the same.

Example – Subjective Human Decisions

Two friends watch the same movie. One finds it "brilliant," and the other finds it "boring." Both are looking at the same movie, yet their decision (opinion) is different because it is shaped by personal taste.

1.2.2 Machine Decision Making – Objective

A computer or machine makes decisions based strictly on **data, rules, logic, and calculations**. Given the same input and the same program, a machine will reach the *same conclusion every time*. This is why machine decision making is called **objective**.

Definition

Objective decision making is decision making based purely on facts, measurable data, and logical rules, free from personal feelings – producing consistent results for the same input.

Basis	**Human Decision Making**	**Machine Decision Making**
Nature	Subjective	Objective
Based on	Emotions, intuition, experience, values	Data, rules, logic, algorithms
Consistency	May vary between individuals/moods	Same input gives the same output
Influenced by	Personal bias, culture, context	Programmed logic and available data

1.2.3 Object Classification: Humans vs. Machines

A good way to understand this difference is to look at how humans and machines **classify objects** (that is, decide what category something belongs to).

How a Human Classifies an Object

When a human sees a fruit and instantly says "that's an apple," the brain is combining years of experience, memory, and even senses like smell and touch – almost instantly and unconsciously. No conscious "calculation" is involved; it feels natural and effortless.

How a Machine Classifies an Object

A machine has no experience or intuition. To classify the same fruit as an "apple," it must:

1. Capture measurable **features** of the object – colour, shape, size, texture, weight.
2. Compare these features, using an **algorithm**, against patterns it has previously learned from data.
3. Calculate which category (label) best matches the features.
4. Output a decision – e.g., "Apple: 96% confidence."

Example – Classifying an Apple

Human: Looks at the fruit, and within a fraction of a second says "Apple" – based on years of seeing apples before. No numbers are consciously involved.

Machine: Measures colour (red/green), roundness, average diameter (7–8 cm), and surface texture from a photo; compares these numbers to thousands of previously labelled fruit images; calculates that this combination of features matches "apple" with high probability, and displays the result.

Quick Recap

- Humans decide subjectively – using emotion, intuition, and experience.
- Machines decide objectively – using data, logic, and algorithms.
- Object classification shows this clearly: humans recognise instantly and unconsciously; machines measure features and calculate a match.

1.3 Machine Learning (ML)

We have seen that machines make objective decisions using data and algorithms. But where do machines get the "rules" for classifying an apple, recognising a face, or predicting the weather? This is where **Machine Learning** comes in.

Definition

Machine Learning (ML) is a branch of Artificial Intelligence in which a computer *learns patterns from data* and improves its performance on a task over time, *without being explicitly programmed with a fixed rule for every single situation.*

Instead of a programmer writing out every possible rule (which would be impossible for something as complex as recognising thousands of different fruit varieties, faces, or handwriting styles), the machine is given large amounts of **data** and learns the underlying patterns on its own.

1.3.1 The Role of Data and Information

Data vs. Information

Data refers to raw, unprocessed facts and figures (numbers, images, text, measurements) that, by themselves, may not mean much.

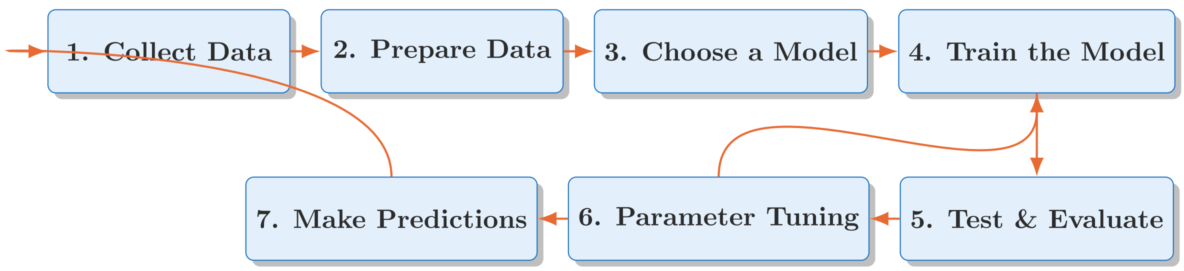
Information is data that has been processed, organised, and given meaning or context, so that it becomes useful for decision making.

Example – Data becoming Information

Data: A list of 500 fruit weights and colours collected from a farm.
Information: "Fruits weighing 150–200g with a red/green skin are most likely apples" – a meaningful pattern extracted from that data.

Machine Learning systems are only as good as the data fed into them. Feeding a machine large amounts of accurate, well-labelled data allows it to identify patterns and turn that raw data into useful information – which then becomes the basis of its objective decisions.

1.3.2 Steps in Machine Learning



1. **Data Collection:** Gathering large amounts of relevant raw data – e.g., thousands of labelled fruit images.

2. **Data Preparation:** Cleaning the data by removing errors, duplicates, or irrelevant information, so it is accurate and usable.

3. **Choosing a Model:** Selecting a suitable learning algorithm/model type best suited to the problem at hand (e.g., a model for classifying images into categories).

4. **Training the Model:** Feeding the prepared data into the chosen **algorithm**, which identifies patterns and relationships between features (e.g., colour, size) and labels (e.g., apple, orange).

5. **Testing & Evaluation:** Checking the trained model against new data it has not seen before, to measure how accurately it makes decisions.
6. **Parameter Tuning:** Adjusting the model's internal settings (parameters) to reduce errors and improve its accuracy, then re-testing it.
7. **Making Predictions:** Once trained, tested, and tuned, the model is used on real, new inputs to make decisions/predictions.

1.3.3 Importance of Programming and Algorithms

Why Algorithms Matter

An **algorithm** is a clear, step-by-step set of instructions used to solve a problem or perform a task. In Machine Learning, algorithms are what allow a computer to *analyse data and find patterns* on its own.

Programming is how these algorithms are translated into a language the computer can actually execute.

Without well-designed algorithms and correct programming:

- A machine cannot process data or identify meaningful patterns.
- A machine cannot "learn" from experience (past data) at all.
- A machine cannot make decisions that appear subjective-like – for example, telling apart very similar-looking fruits the way an experienced human would.

In other words, **good algorithms + good data + good programming** together give a machine the ability to make decisions that come close to human-like judgement, even though the underlying process remains entirely objective and data-driven.

1.3.4 Putting It All Together: The Fruit Sorting Example

Example – An Automated Fruit Sorting Machine

Imagine a machine at a fruit-packing factory that must sort apples, oranges, and bananas moving along a conveyor belt.

1. **Data Collection:** Thousands of images of apples, oranges, and bananas are collected and labelled correctly.
2. **Data Preparation:** Blurry or mislabelled images are removed; all images are resized to a standard format.
3. **Choosing a Model:** A suitable learning algorithm/model, well-suited to classifying images into categories, is selected for the task.
4. **Training the Model:** The chosen model studies the prepared, labelled images and learns which combinations of colour, shape, and size correspond to which fruit.
5. **Testing & Evaluation:** The trained model is shown new fruit images (not used during training) to check how accurately it identifies them.
6. **Parameter Tuning:** The model's settings are adjusted to correct recurring mistakes (e.g., confusing small oranges with lemons), and it is tested again until its accuracy is acceptable.
7. **Making Predictions:** As a fruit passes a camera on the conveyor belt, the tuned model analyses its features in real time and decides: "Orange – send to Bin 2."

Notice that the machine's decision is entirely **objective** – it never "feels" like sorting oranges first, nor does it get "tired" and start guessing. It simply calculates the best match, every single time, based on what it has learned from data.

Quick Recap

- Machine Learning lets a computer learn patterns from data instead of following one fixed rule for every case.
- Data is raw and unprocessed; Information is data that has been organised and given meaning.
- The main steps in ML: collect data, prepare data, train the model, test/evaluate, predict, and improve with feedback.
- Algorithms and programming are essential – without them, a machine cannot learn from data at all.
- The fruit sorting example shows the entire ML pipeline working together, end to end.

Chapter Summary

- **Automated systems** follow fixed rules and are usually deterministic; **autonomous systems** sense, adapt, and decide on their own, and are usually probabilistic.
- **Human decisions** are subjective, shaped by emotion and experience; **machine decisions** are objective, based on data and logic.
- **Machine Learning** allows computers to learn patterns from data through a series of steps – collection, preparation, training, testing, prediction, and improvement – guided always by well-designed **algorithms** and **programming**.

1.4 Solved Example Problems

The topics covered in this chapter are best understood by practising different *types* of problems. Below are several categories of example problems you may be asked to solve, along with worked solutions.

Type 1 – Identifying Automated vs. Autonomous Systems

Problem: State whether each of the following is an Automated or an Autonomous system, giving a reason.

(a) A microwave oven that heats food for a time set by the user.

(b) A warehouse robot that navigates around workers and shelves on its own to deliver packages.

(c) A rice cooker that switches off automatically once the water has boiled away.

Solution:

(a) **Automated** – it simply runs for the fixed time set, following a pre-programmed instruction; it does not sense or adapt.

(b) **Autonomous** – it senses its surroundings (workers, shelves, obstacles) and independently decides its path, adapting as conditions change.

(c) **Automated** – it follows a fixed rule (switch off when a sensor detects a particular temperature/condition); it does not adapt its behaviour beyond this single pre-set trigger.

Type 2 – Identifying Deterministic vs. Probabilistic Behaviour

Problem: Classify the following as Deterministic or Probabilistic, with a reason.

(a) A vending machine that dispenses a fixed item when a particular button is pressed.

(b) A spam filter that marks an email as ”92% likely to be spam.”

Solution:

(a) **Deterministic** – pressing the same button always dispenses the exact same item; there is no uncertainty in the outcome.

(b) **Probabilistic** – the filter expresses its decision as a likelihood (92%) rather than absolute certainty, since it is based on learned patterns from data.

Type 3 – Subjective vs. Objective Decision Making

Problem: For each situation, state whether the decision being made is subjective or objective.

(a) A judge on a cooking show scoring a dish as ”delicious.”

(b) A machine calculating the average marks of a class from a spreadsheet of scores.

Solution:

(a) **Subjective** – taste, and therefore the score, depends on the judge's personal preferences and experience; another judge may score it differently.

(b) **Objective** – the machine follows a fixed mathematical rule (calculating an average) on given data, and will always produce the same result for the same input.

Type 4 – Object Classification: Human vs. Machine

Problem: Ria looks at a photograph and instantly says, ”That's a mango.” A computer program takes the same photograph and, after some processing, also outputs ”Mango: 94% confidence.” Explain how each of them arrived at their answer.
Solution:

Ria (human): Recognises the mango almost instantly and unconsciously, drawing on years of past experience of seeing, touching, and eating mangoes – no conscious calculation is involved.

Computer (machine): Extracts measurable features from the photograph (colour, shape, size, texture), compares them – using an algorithm – with patterns learned from many previously labelled fruit images, and calculates that ”mango” is the best-matching label, expressing the result as a confidence percentage.

Type 5 – Data vs. Information

Problem: A school office has a spreadsheet listing the attendance (present/absent) of every student for every day of the year. State whether this spreadsheet is data or information. Now write one example of the information that could be obtained from it.
Solution:

The raw spreadsheet of daily present/absent entries is **data** – it is unprocessed and, by itself, does not directly tell us anything useful.

Processing it (for example, calculating each student's attendance percentage) turns it into **information**, e.g., ”Rahul has 92% attendance this year,” which is meaningful and can be used for decision making.

Type 6 – Sequencing the Steps of Machine Learning

Problem: The following steps of building a Machine Learning model are jumbled up. Arrange them in the correct order.
Testing the model on unseen data; Selecting a suitable algorithm; Adjusting the model's settings to reduce errors; Gathering thousands of labelled images; Using the finished model to classify a new image; Removing incorrect or duplicate images from the dataset; Feeding the cleaned data to the algorithm so it can learn patterns.
Solution:

1. Gathering thousands of labelled images (*Data Collection*)
2. Removing incorrect or duplicate images from the dataset (*Data Preparation*)
3. Selecting a suitable algorithm (*Choosing a Model*)
4. Feeding the cleaned data to the algorithm so it can learn patterns (*Training the Model*)
5. Testing the model on unseen data (*Test & Evaluate*)
6. Adjusting the model's settings to reduce errors (*Parameter Tuning*)
7. Using the finished model to classify a new image (*Making Predictions*)

Think About It

Try writing one original problem of **Type 1** or **Type 3** yourself, using an example from your own daily life (e.g., a home appliance, or a decision you and a friend disagreed on). Exchange it with a classmate and solve each other's problem.

Exercise – Check Your Understanding

1. Differentiate between Automated and Autonomous systems with one example each.
2. Explain the difference between Deterministic and Probabilistic behaviour.
3. Why is human decision making called subjective while machine decision making is called objective?
4. List and briefly explain the steps involved in Machine Learning.
5. Using the fruit sorting example, explain why algorithms and programming are important for a machine to make decisions.
6. Give one example each of data and the information derived from it.